(i)

$$\begin{aligned} RHS &= \frac{r+1}{r}\left(\frac{1}{\binom{n+r-1}{r}} - \frac{1}{\binom{n+r}{r}}\right) \\ &= \frac{r+1}{r}\left(\frac{r!\,(n-1)!}{(n+r-1)!} - \frac{r!\,n!}{(n+r)!}\right) \\ &= \frac{(r+1)!\,(n-1)!}{r} \cdot \frac{n+r-n}{(n+r)!} \\ &= \frac{(r+1)!\,(n-1)!}{(n+r)!} \\ &= \frac{1}{\binom{n+r}{r+1}} = \text{LHS} \end{aligned}$$

$$\sum_{n=1}^{\infty} \frac{1}{\binom{n+r}{r+1}} = \frac{r+1}{r}\sum_{n=1}^{\infty}\left(\frac{1}{\binom{n+r-1}{r}} - \frac{1}{\binom{n+r}{r}}\right) \lim_{N\to\infty} = \frac{r+1}{r}\left(1 - \frac{1}{\binom{N+r}{r}}\right) = \frac{r+1}{r}$$

When $r = 2$:

$$\sum_{n=1}^{\infty} \frac{1}{\binom{n+2}{3}} = \frac{3}{2}$$

$$\implies \sum_{n=2}^{\infty} \frac{1}{\binom{n+2}{3}} = \frac{3}{2} - \frac{1}{\binom{3}{3}} = \frac{3}{2} - 1 = \frac{1}{2}$$

(ii)

$$\frac{1}{{}^{n+1}C_3} = \frac{(n-2)!\,3!}{(n+1)!} = \frac{3!}{(n+1)n(n-1)}$$

$$(n+1)n(n-1) = n^3 - n < n^3$$

True for $n > 1$. Since $n \geq 3$:

$$\frac{1}{{}^{n+1}C_3} > \frac{3!}{n^3}$$

$$\begin{aligned} \frac{20}{{}^{n+1}C_3} - \frac{1}{{}^{n+2}C_5} - \frac{5!}{n^3} &= \frac{120}{n(n^2-1)} - \frac{120}{n(n^2-1)(n^2-4)} - \frac{120}{n^3} \\ &= \frac{120}{n^3(n^2-1)(n^2-4)}\Big(n^2(n^2-4) - n^2 - (n^2-1)(n^2-4)\Big) \end{aligned}$$

$$(\cdots) = n^4 - 4n^2 - n^2 - (n^4 - 5n^2 + 4) = -480 \cdot \frac{1}{480} \cdot (-480) = -480$$

Since $n \geq 3$, $n^3(n^2-1)(n^2-4) > 0$:

$$\therefore \frac{20}{{}^{n+1}C_3} - \frac{1}{{}^{n+2}C_5} - \frac{5!}{n^3} = \frac{-480}{n^3(n^2-1)(n^2-4)} < 0$$

$$6\sum_{n=3}^{\infty}\frac{1}{n^3} < \sum_{n=3}^{\infty}\frac{1}{{}^{n+1}C_3} = \sum_{n=2}^{\infty}\frac{1}{{}^{n+2}C_3} = \frac{1}{2}$$

$$\implies \sum_{n=3}^{\infty}\frac{1}{n^3} < \frac{1}{12}$$

Left inequality:

$$6\sum_{n=3}^{\infty}\frac{1}{n^3} < \sum_{n=3}^{\infty}\frac{1}{{}^{n+1}C_3} = \sum_{n=2}^{\infty}\frac{1}{{}^{n+2}C_3} = \frac{1}{2}$$

$$\therefore \sum_{n=1}^{\infty}\frac{3!}{n^3} < \frac{3!}{1} + \frac{3!}{8} + \frac{1}{2} = 6 + \frac{3}{4} + \frac{1}{2} = \frac{29}{4}$$

$$\implies \sum_{n=1}^{\infty}\frac{1}{n^3} < \frac{29}{24} = \frac{116}{96}$$

Right inequality:

$$\sum_{n=3}^{\infty}\left(\frac{20}{{}^{n+1}C_3} - \frac{1}{{}^{n+2}C_5}\right) < \sum_{n=3}^{\infty}\frac{5!}{n^3}$$

From Part (i), when $r = 4$: $\displaystyle\sum_{n=1}^{\infty}\frac{1}{{}^{n+4}C_5} = \frac{5}{4}$

$$\therefore \sum_{n=3}^{\infty}\left(\frac{20}{{}^{n+1}C_3} - \frac{1}{{}^{n+2}C_5}\right) = 20 \times \frac{1}{2} - \sum_{n=1}^{\infty}\frac{1}{{}^{n+4}C_5} = 10 - \frac{5}{4} = \frac{35}{4}$$

$$\implies \sum_{n=3}^{\infty}\frac{5!}{n^3} > \frac{35}{4}$$

$$\implies \sum_{n=1}^{\infty}\frac{1}{n^3} > \frac{35}{4 \times 5!} + \frac{1}{1} + \frac{1}{8} = \frac{7}{96} + \frac{96}{96} + \frac{12}{96} = \frac{115}{96}$$

$$\therefore \boldsymbol{\frac{115}{96} < \sum_{n=1}^{\infty}\frac{1}{n^3} < \frac{116}{96}}$$

This solution was submitted by **Jon**.

Published and refined by **Jon**.

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